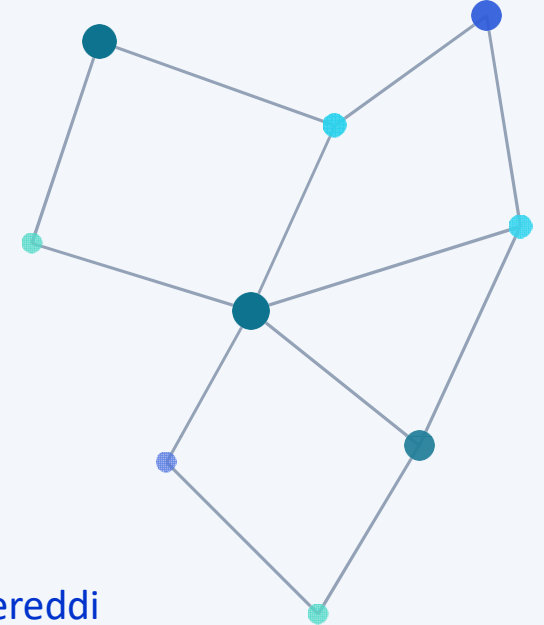




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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING •
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Explainable AI for Human Resource Analytics using Kolmogorov–Arnold Networks



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The HR Attrition Challenge

Human Resource Management plays an important role in

- Talent acquisition
- Employee retention
- Organizational productivity
- Financial sustainability

One of the most **challenging** HR tasks is

- Predicting which employees are likely to leave the organization.

Class Imbalance

Attrition cases are a small minority (~16%). Models that optimise for overall accuracy silently default to predicting stays. [7, 8]

Black-Box Models

Deep networks rarely explain their predictions. SHAP and LIME help, but are approximate, post-hoc, and unstable. [9, 10]

Related Work: From Statistics to XAI

1

Statistical Methods

Linear regression, survival analysis — interpretable but limited to linear effects

2

Machine Learning

Decision trees, SVM, Random Forest, XGBoost — better accuracy, limited transparency

3

Deep Learning

Captures complex patterns; imbalance-sensitive and black-box [6, 9]

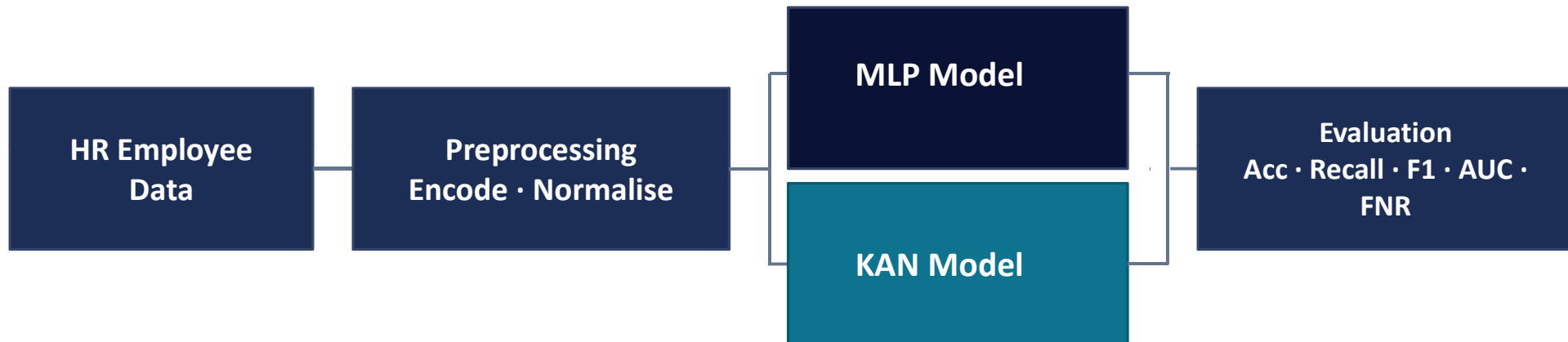
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Explainable AI

SHAP / LIME post-hoc — approximate. KANs underexplored in HR — our focus [12]

Each era added predictive power at the cost of transparency — motivating an inherently interpretable deep learning architecture.

Proposed Approach: System Pipeline



Identical data splits, optimizer, and class weights for both models — ensuring a fair comparison.

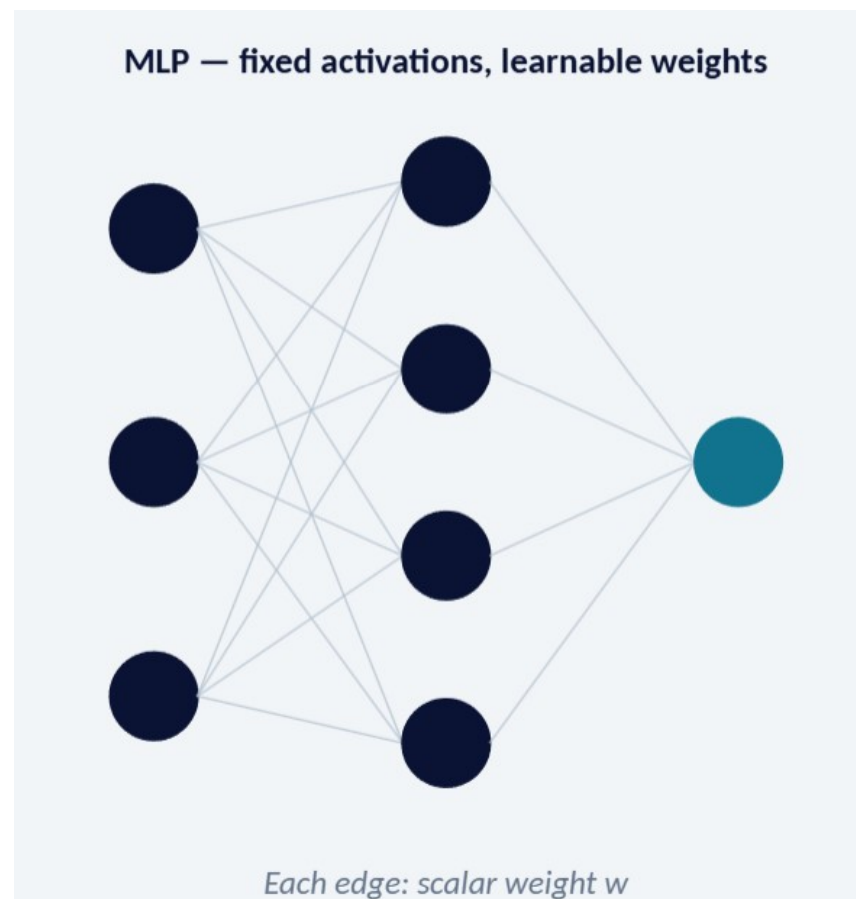
Multi-Layer Perceptron (MLP)

- ❑ Fully connected feedforward network: input layer → hidden layers → output [11]
- ❑ Each node applies a fixed non-linear activation (e.g. ReLU) to a learned linear combination of inputs
- ❑ Trained end-to-end via backpropagation to minimise loss
- ❑ **Limitation:** feature influence is entangled across layers — not directly interpretable

Equations:

$$h^{(l)} = \sigma(W^{(l)}h^{(l-1)} + b^{(l)}), l = 1, \dots, L-1 \text{ and } \hat{y} = W^{(L)}h^{(L-1)} + b^{(L)}$$

Diagram: MLP network — fixed activations, learnable weights.
Each edge carries a single learnable scalar weight w .



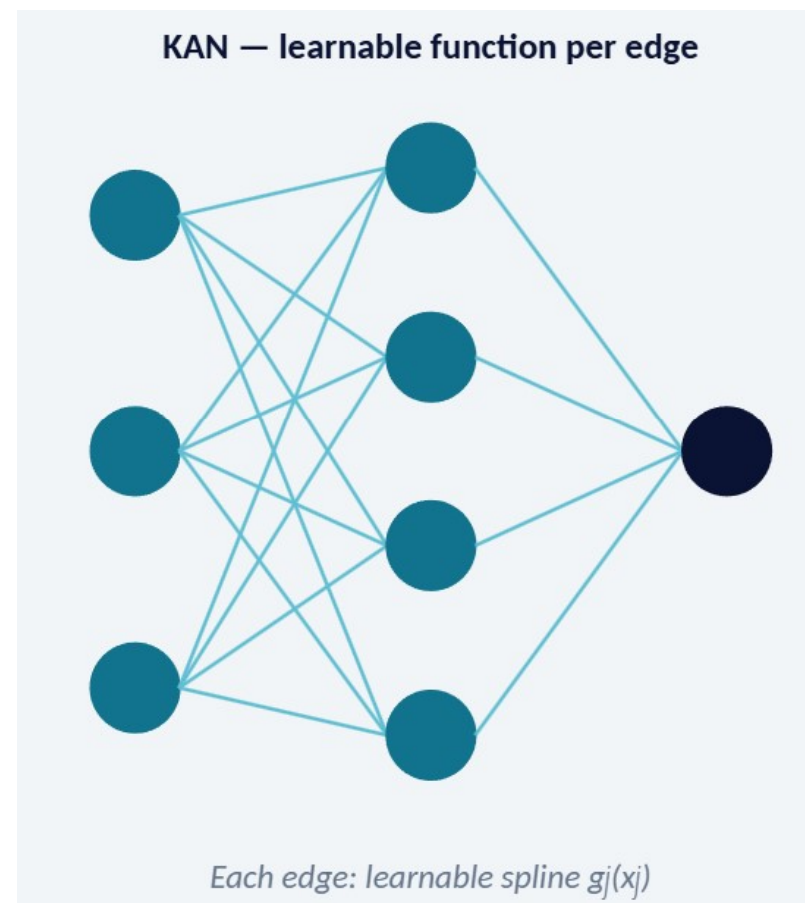
Identical data splits, optimizer, and class weights for both models — ensuring a fair comparison.

Kolmogorov-Arnold Networks (KAN)

- ❑ Grounded in the Kolmogorov–Arnold theorem: any multivariate function can be expressed as sums of univariate functions [12]
- ❑ A learnable non-linear function g_j replaces each scalar weight on every edge
- ❑ The additive decomposition makes every feature's contribution directly attributable
- ❑ Interpretability is structural — no post-hoc explanation step needed

Equations: $f(x) = \sum_j g_j(x_j)$ and $\hat{y} = \sigma(\sum_j g_j(x_j) + b)$

Diagram: *KAN network — learnable function per edge. Each edge learns its own spline function $g_j(x_j)$.*



Identical data splits, optimizer, and class weights for both models — ensuring a fair comparison.

Handling Class Imbalance & Interpretability

Class-Weighted Loss

$$\mathcal{L}_{WBCE} = -\frac{1}{N} \sum_{i=1}^N [w_1 y_i \log(\hat{y}_i) + w_0 (1 - y_i) \log(1 - \hat{y}_i)]$$

- w_1 and w_0 are inversely proportional to class frequency
- False negatives (missed attrition) are penalised more heavily, directly boosting recall for the minority class

Inherent Interpretability

- MLP: feature influence is distributed across all hidden layers — SHAP or LIME needed as an extra, approximate step
- KAN: $g_j(x_j)$ directly quantifies each feature's contribution to the final prediction
- No post-hoc step required — interpretability is structural, not bolted on

Dataset & Experimental Setup

IBM HR Analytics Dataset



16.1% Attrition (minority class)

83.9% Retained employees

A realistic depiction of HR data — posing a real risk of high false negatives.

Features & Preprocessing

Age, Gender, Department, Job Role, Monthly Income, Job Level, Years at Company, Years Since Last Promotion, OverTime, Work–Life Balance, Job Satisfaction, Performance Rating

- ❑ Categorical attributes encoded; numerical attributes normalised
- ❑ Stratified 80/20 train–test split to preserve class ratio
- ❑ Both models use identical splits, loss function, optimizer (Adam), and class weights — ensuring a fair comparison

Evaluation Metrics

- ❑ **Accuracy** — Proportion of all correctly classified instances
- ❑ **Precision** — Of predicted attrition cases, the fraction that truly left
- ❑ **Recall** — Of actual attrition cases, the fraction correctly identified
- ❑ **F1-score** — Harmonic mean of Precision and Recall
- ❑ **Specificity** — Correctly identified non-attrition employees
- ❑ **FNR** — Fraction of attrition cases missed (False Negative Rate)
- ❑ **AUC** — Discriminative power across all classification thresholds (Area Under Curve)

ROC and Precision–Recall curves provide threshold-independent evaluation. AUC confidence intervals computed via 1000-sample bootstrapping.

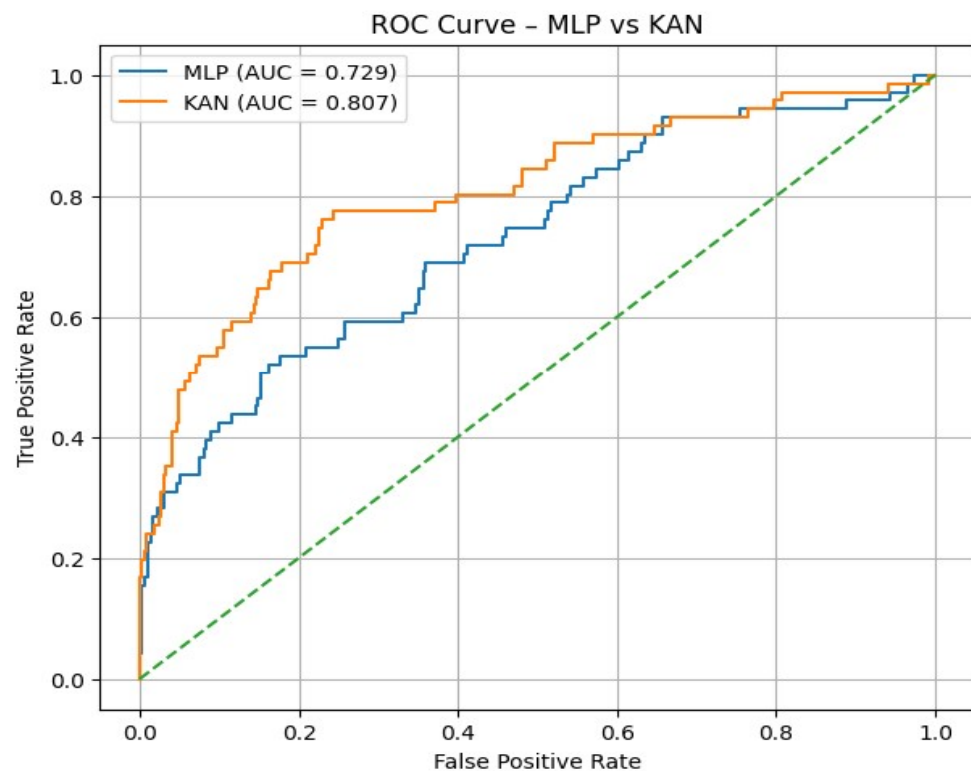
Results: MLP vs KAN Performance

Metric	MLP	KAN
Accuracy	0.82	0.79
Precision (Attrition)	0.55	0.71 ✓
Recall (Attrition)	0.41	0.76 ✓
F1-score	0.47	0.73 ✓
Specificity	0.90	0.83
False Negative Rate (FNR)	0.59	0.24 ✓
AUC	0.68	0.81 ✓

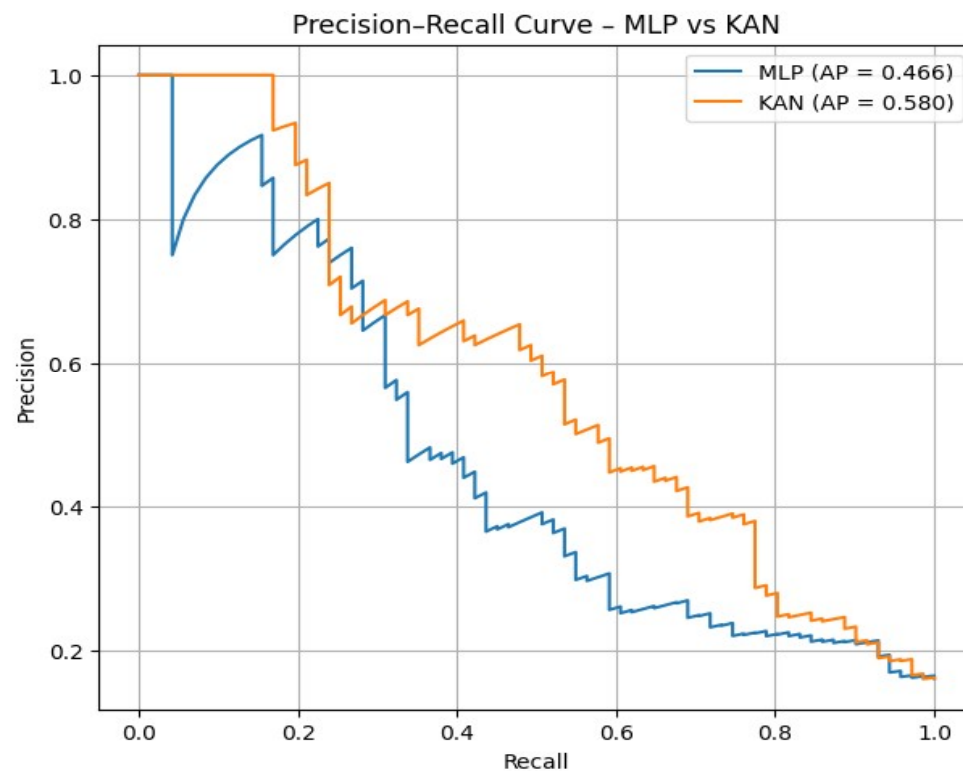
FNR: 0.59 → 0.24

AUC: 0.68 → 0.81

ROC & Precision-Recall Curves



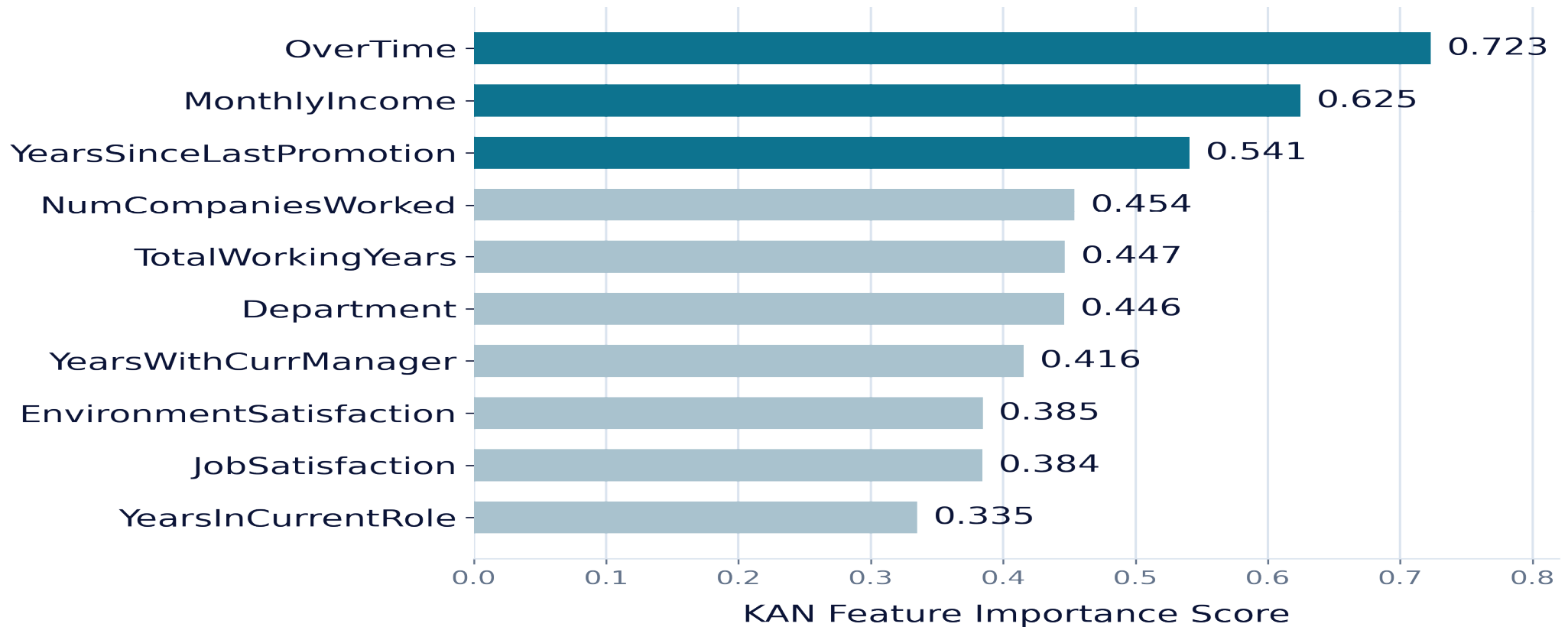
AUC: KAN 0.81 vs MLP 0.68 — KAN dominates across all thresholds



KAN sustains higher precision across a wider range of recall values

KAN Feature Importance

OverTime, Monthly Income, and Years Since Last Promotion emerge as the strongest attrition drivers — consistent with established HR knowledge.



From Predictions to HR Action

1

Proactive Identification

Higher recall surfaces at-risk employees earlier, before they disengage — enabling pre-emptive intervention.

2

Targeted Interventions

Compensation adjustments, promotions, and workload restructuring guided by real, ranked feature drivers.

3

Transparent Policy-Making

Feature-level evidence builds organisational trust in data-driven HR decisions — without a black-box caveat.

Conclusions & Future Directions

Conclusions

- KAN outperforms MLP in recall, F1-score, AUC, and robustness under class imbalance
- False negatives fall sharply — FNR drops from 0.59 to 0.24
- Feature-additive modelling gives intrinsic interpretability, without post-hoc tools
- Key attrition drivers: overtime, compensation, promotion history, and work–life balance

Future Directions

- Extend to longitudinal employee data over longer durations
- Validate on larger, more diverse organisational datasets
- Explore medical diagnostics, where class imbalance is common — though clinical trust remains paramount

Key References

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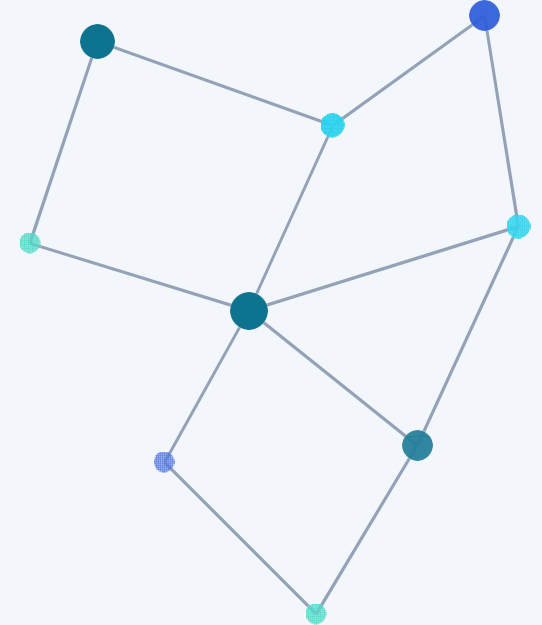


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Thank You

Questions & Discussion



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